Proof of Dislocation Energy in FCC: [100] vs [110] Directions

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> *7. For a metal with FCC structure, E_1 and E_2 are the energies necessary to generate dislocations in the directions $[100]$ and $[110]$, respectively. Demonstrate that $E_1/E_2 = 2$.* *(Solution: theory)*
>
> Solution: theory

PHASE 1: HYPOTHESIS & PARAMETERS

Lattice Model:

- Structure: Face-Centred Cubic (FCC), with lattice parameter a.
- Relation to the atomic radius: $a \quad 2 = 4R \implies a = 2 \quad 2R$.

Definition of Energies:

- E_1: Elastic energy per unit length to generate a dislocation with Burgers vector along the $[100]$ direction.
- E_2: Elastic energy per unit length to generate a dislocation with Burgers vector along the $[110]$ direction.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing Frank's theory and the mechanics of dislocations [Session 4 Slides 29, 36]:

1. **Frank's Rule for the Elastic Energy of a Dislocation:**

The elastic strain energy stored in the elastic distortion field surrounding a dislocation is directly proportional to the **square of the magnitude of its Burgers vector**:

$$E \propto |b|^2$$

2. **Determination of the Minimum Translation Vectors:**

For a dislocation to be kinematically stable in the crystal lattice, its Burgers vector must connect two equivalent lattice atomic positions:

- In the $[100]$ direction, the minimum translation joining two FCC lattice nodes (two adjacent vertices) is the full edge:

$$b_1 = a[1\,0\,0]$$

- In the $[110]$ direction, owing to the presence of the face-centred atom, the shortest lattice translation goes from a vertex to the face centre:

$$b_2 = \frac{a}{2}[1\,1\,0]$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

Study notes written from the official course material (Problems T2_CrystStruct.pdf; Problems T2 defects.pdf).
Online version: subjects/aerospace-materials-1/problemas/topic-2-problems.html